



Intestinal Protozoa — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which protozoan is the only one found in the lumen of the human small intestine?

EASY

- A. *Cryptosporidium parvum*
- B. *Giardia lamblia*
- C. *Balantidium coli*
- D. *Entamoeba histolytica*
- E. *Trichomonas vaginalis*

2. Which protozoan is the only ciliate infecting humans?

EASY

- A. *Giardia lamblia*
- B. *Cryptosporidium parvum*
- C. *Entamoeba coli*
- D. *Balantidium coli*
- E. *Plasmodium*

3. Which stain is the method of choice for detecting *Cryptosporidium* oocysts in stool?

EASY

- A. Gram stain
- B. Iodine stain
- C. Giemsa
- D. India ink
- E. Modified acid-fast stain

4. What is the infective form of *Giardia*?

EASY

- A. Mature cyst with 4 nuclei
- B. Trophozoite
- C. Immature cyst with 2 nuclei
- D. Oocyst
- E. Ciliated trophozoite

5. Which motility is characteristic of the *Giardia* trophozoite?

MEDIUM

- A. Slow "falling leaf" movement
- B. Rapid ciliary rotation
- C. Amoeboid pseudopods
- D. Undulating membrane
- E. Gliding

6. Which stool description fits giardiasis?

MEDIUM

- A. Pale, foul, fatty stools that may float, with mucus but usually no blood
- B. Bloody mucoid stool with pus
- C. Rice-water stool
- D. Black tarry stool
- E. Watery stool with red acid-fast spheres only

7. What is the approximate infective dose of *Giardia*?

MEDIUM

- A. 1–2 cysts
- B. 10^3 cysts
- C. 10–100 cysts
- D. 10^5 – 10^8 cysts
- E. 10^9 cysts

8. How does *Giardia* injure the small intestine?

MEDIUM

- A. Invades the submucosa and forms abscesses
- B. Produces cholera-like toxin
- C. Attaches by a sucking disc, causing villous damage without tissue invasion
- D. Penetrates the liver
- E. Lives inside enterocytes

9. A child with chronic diarrhea needs an alternative to metronidazole. Which options are listed?

MEDIUM

- A. Tetracycline or doxycycline
- B. Praziquantel
- C. Mebendazole
- D. Furazolidone or nitazoxanide
- E. Albendazole only

10. A pregnant woman has symptomatic giardiasis. Which drug is an option?

MEDIUM

- A. Tetracycline
- B. Furazolidone
- C. Doxycycline
- D. Tinidazole only
- E. Paromomycin

11. Which feature describes *Cryptosporidium* oocysts?

MEDIUM

- A. Oval 12 × 8 µm with 4 nuclei
- B. Spherical 40–60 µm with a macronucleus
- C. Planoconvex 50–60 µm
- D. Bile-stained with a mammillated coat
- E. Acid-fast, iodine-negative, ~5 µm, with 4 crescent-shaped sporozoites

12. Which statement about cryptosporidiosis in AIDS is correct?

MEDIUM

- A. It is usually asymptomatic
- B. It causes chronic, persistent, profuse diarrhea with dehydration and weight loss
- C. It causes bloody dysentery
- D. It causes fatty stool only
- E. It requires tetracycline

13. Which set describes the *Balantidium* trophozoite?

MEDIUM

- A. 15 × 9 µm with 2 nuclei and a sucking disc
- B. 5 µm acid-fast sphere
- C. 12 × 8 µm oval cyst
- D. Crescent-shaped sporozoite
- E. 60–70 µm, ovoid, kidney-shaped macronucleus and a micronucleus, invasive

14. Which intestinal protozoan is a zoonosis with pigs, monkeys and rats as reservoirs and humans as accidental hosts?

MEDIUM

- A. *Giardia lamblia*
- B. *Balantidium coli*
- C. *Cryptosporidium hominis*
- D. *Enterobius vermicularis*
- E. *Ancylostoma duodenale*

15. Which oocyst type causes autoinfection in cryptosporidiosis?

HARD

- A. Thick-walled oocysts
- B. Iodine-stained oocysts
- C. Unsporulated cysts
- D. Thin-walled oocysts (~20%)
- E. Bile-stained oocysts

16. Balantidiasis resembles amoebiasis. What differentiates it from *E. histolytica* in the lecture?

HARD

- A. It never causes dysentery
- B. It affects the small intestine
- C. Extraintestinal invasion is uncommon
- D. It occurs only in HIV
- E. Its cyst has 4 nuclei

17. Two people share a house with pigs. One is a healthy adult, the other a malnourished alcoholic. Only the alcoholic has dysentery and colonic ulcers. Why?

HARD

- A. Only alcoholics can be infected
- B. Pigs infect only malnourished people
- C. *B. coli* is usually commensal; disease follows malnutrition, alcoholism or reduced resistance
- D. Cysts cannot infect healthy adults
- E. The healthy adult is immune to all protozoa

18. A stool antigen test for *Giardia* is positive, whereas another patient has only *Giardia* antibodies. What is the correct interpretation?

HARD

- A. Antibodies indicate active infection
- B. Antigen indicates active infection; antibodies are mainly epidemiological
- C. Neither has diagnostic value
- D. Antigen is only for research
- E. Both indicate cure

19. A patient with profuse persistent diarrhea and red spheres in stool has no effective specific therapy for the organism. What is the appropriate management?

HARD

- A. Nitazoxanide plus rehydration, electrolytes and nutrients
- B. Metronidazole alone
- C. Tetracycline
- D. Paromomycin alone
- E. Antimotility drugs alone

20. A patient with diarrhea shows cysts with hyaline walls and 4 nuclei in stool; another shows a huge ciliated organism. Which pair is correct?

HARD

- A. Balantidium; *Giardia*
- B. *Cryptosporidium*; *Giardia*
- C. *Giardia*; *Cryptosporidium*
- D. *Giardia*; Balantidium
- E. Balantidium; *Cryptosporidium*

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	G. lamblia lives in the duodenum and upper jejunum.
2	D	EASY	B. coli belongs to the phylum Ciliophora and is the largest human protozoan.
3	E	EASY	Oocysts appear as red spheres and do not stain with iodine.
4	A	EASY	Cysts survive in water; mature ones have 4 nuclei.
5	A	MEDIUM	Falling-leaf motility, sucking disc and 4 pairs of flagella identify it.
6	A	MEDIUM	Giardiasis causes steatorrhea; it does not invade tissue.
7	C	MEDIUM	Low infective dose explains easy water-borne spread.
8	C	MEDIUM	It causes villous damage, apoptosis and lymphocytic infiltration without invasion.
9	D	MEDIUM	Furazolidone or nitazoxanide are alternatives, particularly in children.
10	E	MEDIUM	Paromomycin is the option listed in pregnancy.
11	E	MEDIUM	Oocysts also have thick-walled and thin-walled forms.
12	B	MEDIUM	Immunocompromised patients develop severe persistent diarrhea; immunocompetent patients usually self-limit.
13	E	MEDIUM	The trophozoite is the motile invasive stage found in dysenteric stool.
14	B	MEDIUM	B. coli completes its life cycle in one host (monoxenous); human beings acquire cysts from pigs, animals or carriers.
15	D	HARD	Thick-walled oocysts resist the environment; thin-walled ones autoinfect.
16	C	HARD	B. coli invades the colonic mucosa but rarely spreads outside the intestine.
17	C	HARD	Disease is more likely with reduced host resistance.
18	B	HARD	Stool ELISA / IFT antigen detects active infection.
19	A	HARD	Supportive treatment is central; nitazoxanide is listed.
20	D	HARD	Giardia cysts are oval with 4 nuclei; Balantidium is a large ciliate.